Learning Global Probabilistic Explanations

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Abstract

Understanding predictions made by complex black-box models remains a significant challenge in explainable artificial intelligence. While local explanations provide detailed insights for individual instances, there is a persistent need for global explanations that capture a model's overall decision-making behavior across the entire input space. In this paper, we introduce a principled framework for constructing *global probabilistic explanations* for black-box classifiers. We define a candidate explanation as any subset of features and measure its *relevance* by the probability that the classifier assigns identical labels to pairs of instances that agree on the selected features. For a classifier represented by an arbitrary Boolean function f, a distribution over instance pairs, and a size constraint k, our objective is to identify a subset of features K (of size at most k) that maximizes relevance. Although this stochastic constraint optimization task is generally computationally intractable, we show that, under the uniform distribution, near-optimal explanations can be efficiently computed using membership query access to f and Fourier-analytic techniques. By leveraging agnostic PAC-learning methods for junta functions, our algorithm achieves fixed-parameter tractability in k. Empirical results on real-world datasets demonstrate that our approach attains higher relevance and runtime efficiency than state-of-the-art benchmarks.

1 Introduction

The classification problem has long served as a foundation of machine learning research [16]. The field has evolved dramatically, advancing from early linear classifiers and decision trees to contemporary deep neural networks, ensemble approaches, and graph neural networks [15]. While predictive accuracy was once the dominant concern, there is now heightened recognition of the need for interpretability and explainability—especially as machine learning models are increasingly deployed in consequential, real-world settings [11]. In these domains, high predictive performance alone is inadequate: stakeholders demand transparency and well-justified decisions. This necessity has fueled rapid progress in *explainable artificial intelligence (XAI)*, which aims to decipher complex models, by making their predictions transparent and accessible to human users [13].

Within XAI, *formal explainability* [7, 12] stands out by offering rigorous, theoretically grounded guarantees for explanations. A key challenge is to clarify *why* a particular instance $\boldsymbol{x}$, described by d features, is assigned a specific label $f(\boldsymbol{x})$ by a classifier f [5, 17]. Such explanations are typically constructed by identifying a subset of features K so that altering features outside K does not affect the prediction. In other words, the projection of $\boldsymbol{x}$ onto K, denoted $\boldsymbol{x}_K$, suffices to determine $f(\boldsymbol{x})$. Accordingly, K is referred to as an *abductive explanation* [5] or *sufficient reason* [17].

Although abductive explanations are guaranteed to be sound, they are often too large and complex for effective human interpretation. To address this limitation, recent research has increasingly focused on (abductive) *probabilistic* explanations, which aim to strike a balance between interpretability and fidelity by identifying succinct feature subsets [8, 18]. In this approach, the explanatory power of a

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subset K is quantified by its *local relevance* with respect to a classifier f and a reference instance $\boldsymbol{x}$. Specifically, local relevance is defined as the probability that f assigns the same label to a random instance $\boldsymbol{z}$ as to $\boldsymbol{x}$, assuming both instances share identical values for features in K. Given a size constraint k and an instance distribution $\mathcal{D}$, the objective is to find a subset K of size at most k that minimizes the local relevance error:

$$\mathbb{P}_{\boldsymbol{z}\sim\mathcal{D}}[f(\boldsymbol{x}) \neq f(\boldsymbol{z}) \mid \boldsymbol{x}_K = \boldsymbol{z}_K] \tag{1}$$

While much prior work has focused on the *local* perspective—explaining individual predictions—recent advances have embraced a more ambitious *global* perspective, aiming to capture a model's behavior across the entire input space [1, 2]. In this setting, the feature subset K acts as a *junta* [3, 4], partitioning the input space into regions defined by all possible assignments to K. With a size constraint k and a distribution $\mathcal{P}$ over *instance pairs*, the goal is to identify a subset K of size at most k that minimizes the global relevance error:

$$\mathbb{P}_{(\boldsymbol{x},\boldsymbol{z})\sim\mathcal{P}}[f(\boldsymbol{x}) \neq f(\boldsymbol{z}) \mid \boldsymbol{x}_K = \boldsymbol{z}_K] \tag{2}$$

In contrast to approaches that define global relevance as the average of local errors [2], our formulation introduces a symmetric metric that measures the functional consistency of the partition induced by K. By evaluating the probability of agreement over the full distribution of instance pairs $\mathcal{P}$, we emphasize explanations that yield internally homogeneous regions within the input space—a property that enhances semantic interpretability and aligns well with Fourier-analytic methods [10, 14].

However, when $\mathcal{P}$ is an arbitrary distribution over instance pairs and f is a neural network, solving the stochastic optimization problem (2) is highly intractable. Indeed, it subsumes the local explanation problem (1), which is $\mathsf{NP}^{\mathsf{PP}}$-hard [18]. This computational hardness raises the question of whether efficient algorithms can be developed for global explanations under restricted distributions.

We answer this question affirmatively: when $\mathcal{P}$ is the uniform distribution over instance pairs—even if f is an arbitrary Boolean function available only via membership queries—near-optimal global explanations can be computed efficiently using Fourier-analytic methods. We provide a structural characterization of optimal explanations via the Fourier spectrum of f and propose an agnostic learning algorithm that integrates the Kushilevitz-Mansour (KM) technique [9] for identifying influential features with the Gopalan-Kalai-Kannan (GKK) pruning strategy [6] for efficiently selecting a subset K of size at most k. Our algorithm is fixed-parameter tractable in k and operates in time polynomial in d and 2^k. Extensive experiments on real-world datasets demonstrate that our approach consistently outperforms state-of-the-art methods in both relevance and computational efficiency.

2 Computing Probabilistic Explanations

Probabilistic explanation problems are defined over two classes of objects: models and distributions. In this paper, the class of models is a parameterized family $\mathcal{F} = \bigcup_{d \geq 1} \mathcal{F}_d$ of Boolean functions. Each model $f \in \mathcal{F}_d$ is a function $f : \{0,1\}^d \to \{-1,+1\}$ that maps data instances over $[d] = \{1, \ldots, d\}$ interpretable features to binary labels. The second class is a parameterized family $\mathcal{D} = \bigcup_{d \geq 1} \mathcal{D}_d$ of distributions over instance pairs, which we use to evaluate the performance of explanations. A distribution $\mathcal{D}_d$ is called *local* if it is supported on pairs of the form $(\boldsymbol{x}, \boldsymbol{z})$ where $\boldsymbol{x}$ is a fixed reference instance and $\boldsymbol{z}$ is drawn from some distribution over $\{0,1\}^d$. In contrast, $\mathcal{D}_d$ is called *global* if it is supported on arbitrary pairs of instances from $\{0,1\}^d$.

For a model $f \in \mathcal{F}_d$ and a distribution $\mathcal{D}_d$, an *explanation* is a subset of features $K \subseteq [d]$. By $\boldsymbol{x}_K$, we denote the restriction of a data instance $\boldsymbol{x}$ to the features in K. The explanation K is called *local* if $\mathcal{D}_d$ is local, and *global* if $\mathcal{D}_d$ is global. The *relevance* of K is defined as the probability that f assigns the same label to two instances drawn from $\mathcal{D}_d$, conditioned on their agreement on the features in K. The corresponding relevance error function is given by:

$$\mathsf{err}_{f,\mathcal{D}_d}(K) = \mathbb{P}_{(\boldsymbol{x},\boldsymbol{z}) \sim \mathcal{D}_d}[f(\boldsymbol{x}) \neq f(\boldsymbol{z}) \mid \boldsymbol{x}_K = \boldsymbol{z}_K]$$

Two optimization tasks naturally arise from this definition. One can fix a threshold $\tau \in [0,1]$ and ask if there exists an explanation K of size at most k such that $\mathsf{err}_{f,\mathcal{D}_d}(K) \leq 1 - \tau$. Alternatively, one can seek to find an explanation K of size at most k that minimizes $\mathsf{err}_{f,\mathcal{D}_d}(K)$. In this paper, we focus on the latter optimization problem, for which the decision version is defined as follows.

k-SIZE τ-RELEVANT EXPLANATION$[\mathcal{F}, \mathcal{D}]$
Input: A model $f \in \mathcal{F}_d$, a distribution $\mathcal{D}_d$, an integer $k \leq d$, and a threshold $\tau \in [0,1]$.
Question: Does there exist a subset $K \subseteq [d]$ of size at most k such that $\mathsf{err}_{f,\mathcal{D}_d}(K) \leq 1 - \tau$?

The next result, adapted from [18], establishes that the RELEVANT EXPLANATION problem is highly intractable when the models are neural networks and the distributions are local. Specifically, let $\mathcal{TN}$ denote the class of Boolean functions representable by threshold networks with weights $\{-1, 0, +1\}$ and arbitrary integer biases. We note that $\mathcal{TN}$ is expressive enough to efficiently represent the class of polynomial-sized Boolean circuits (the class P/poly) defined over the gates AND, OR, NOT, and XOR. In addition, let $\mathcal{LU}$ be the class of *local uniform* distributions, that is, $\mathcal{LU}_d = \{\delta_{\boldsymbol{x}} \times \mathcal{U}_d : \boldsymbol{x} \in \{0,1\}^d\}$, where $\delta_{\boldsymbol{x}}$ is the point mass centered at $\boldsymbol{x}$ and $\mathcal{U}_d$ is the uniform distribution over $\{0,1\}^d$.

Theorem 1 ([18])**.** k-SIZE τ-RELEVANT EXPLANATION [$\mathcal{TN}$, $\mathcal{LU}$] is $\mathsf{NP}^{\mathsf{PP}}$-hard.

Intuitively, this result indicates that finding a τ-relevant explanation is confronted with two independent sources of complexity: the combinatorial search for K and the difficulty of evaluating $\mathsf{err}_{f,\mathcal{D}_d}(K)$. A natural way to alleviate the first issue is to consider fixed-parameter tractability with respect to k. Unfortunately, even if we remove the evaluation complexity by setting $\tau = 1$, the problem remains hard even for conceptually simpler monotone models. Specifically, let $\mathcal{MTN}$ denote the class of Boolean functions representable by monotone threshold networks with weights $\{0, +1\}$ and biases in $\{-1, 0\}$.

Theorem 2. k-SIZE 1-RELEVANT EXPLANATION [$\mathcal{MTN}$, $\mathcal{LU}$] is $W[2]$-hard with respect to the parameter k.

Proof. We perform a reduction from the HITTING SET problem. Given a universe $U = \{1, \ldots, n\}$, a collection of subsets $\mathcal{S} = \{S_1, \ldots, S_m\}$ where $S_j \subseteq U$, and an integer k, we ask if there exists $H \subseteq U$ such that $|H| \leq k$ and $H \cap S_j \neq \varnothing$ for all $j \in [m]$. This problem is $W[2]$-complete.

Construct an instance $(f, \mathcal{D}_d, k, 1)$ where $d = n$. We define the model $f \in \mathcal{MTN}_d$ as the monotone Boolean function $f(\boldsymbol{z}) = \bigwedge_{j=1}^{m} \left(\bigvee_{i \in S_j} z_i \right)$. This function can be implemented by a 2-layer monotone threshold network where the first layer computes the OR clauses (biases $b = 0$) and the second layer computes the conjunction (AND) of these results (bias $b = -(m-1)$). We set the local distribution $\mathcal{LU}$ centered at the all-ones vector $\boldsymbol{x} = \mathbf{1}$. Note that $f(\boldsymbol{x}) = 1$.

A subset $K \subseteq [d]$ is a 1-relevant explanation if and only if $\mathsf{err}_{f,\mathcal{D}_d}(K) = 0$, which implies $f(\boldsymbol{z}) = f(\boldsymbol{x}) = 1$ for all $\boldsymbol{z}$ such that $\boldsymbol{z}_K = \boldsymbol{x}_K = \mathbf{1}_K$. If K is a hitting set for $\mathcal{S}$, then for any $j \in [m]$, there exists $i \in S_j \cap K$. Thus, $z_i = 1$, ensuring the j-th clause is satisfied. Consequently, $f(\boldsymbol{z}) = 1$ for all such $\boldsymbol{z}$, and $\mathsf{err}(K) = 0$. Conversely, if K is not a hitting set, there exists a clause S_j such that $S_j \cap K = \varnothing$. Setting $z_i = 1$ for $i \in K$ and $z_i = 0$ for $i \notin K$ yields $f(\boldsymbol{z}) = 0$. Since $\mathcal{U}_d$ has full support, $\mathbb{P}[f(\boldsymbol{z}) \neq 1 \mid \boldsymbol{z}_K = \mathbf{1}_K] > 0$, and thus $\mathsf{err}(K) > 0$. Therefore, a hitting set of size k exists if and only if there is a k-size 1-relevant explanation. □

As highlighted in the above results, the problem of computing probabilistic explanations is intractable when $\mathcal{D}$ is the product of a Dirac distribution and the uniform distribution. In this asymmetric setting, $\delta_{\boldsymbol{x}}$ concentrates all probability mass on a single instance $\boldsymbol{x}$, causing the optimization to focus exclusively on finding a feature subset K tailored to $\boldsymbol{x}$'s classification relative to the entire input space. In the rest of this study, we shall turn our attention to the more natural case where $\mathcal{D}$ is the uniform distribution $\mathcal{U} \times \mathcal{U}$ over instance pairs. In this symmetric scenario, f can be analyzed as a function on the whole hypercube, enabling us to exploit its global Fourier spectrum for efficient, near-optimal explanations.

3 Learning Global Probabilistic Explanations

The core component of Fourier analysis is an orthonormal basis of functions with respect to the underlying probability distribution [14]. For the uniform distribution $\mathcal{U}$ over $\{0,1\}^d$, the Fourier basis is formed by the parity functions $\phi_S(\boldsymbol{x}) = (-1)^{\sum_{i \in S} x_i}$ for all $S \subseteq [d]$. Any pseudo-Boolean function $f : \{0,1\}^d \to \mathbb{R}$ admits a unique Fourier expansion:

$$f(\boldsymbol{x}) = \sum_{S \subseteq [d]} \hat{f}(S)\phi_S(\boldsymbol{x}),$$

where the coefficients are given by $\hat{f}(S) = \mathbb{E}_{\boldsymbol{x} \sim \mathcal{U}}[f(\boldsymbol{x})\phi_S(\boldsymbol{x})]$. By Plancherel's theorem, the inner product of two pseudo-Boolean functions f and g is $\langle f, g \rangle = \mathbb{E}_{\boldsymbol{x}}[f(\boldsymbol{x})g(\boldsymbol{x})] = \sum_S \hat{f}(S)\hat{g}(S)$. For a positive integer p, the *functional* and *spectral* p-norms of f are respectively defined as:

$$\|f\|_p = (\mathbb{E}_{\boldsymbol{x} \sim \mathcal{U}}[|f(\boldsymbol{x})|^p])^{1/p} \quad \text{and} \quad L_p(f) = \left(\sum_{S \subseteq [d]} |\hat{f}(S)|^p \right)^{1/p}.$$

By Parseval's identity, $\|f\|_2 = L_2(f)$, which is equal to one when f is Boolean. As special cases, $L_\infty(f) = \max_{S \subseteq [d]} |\hat{f}(S)|$ and $L_0(f) = |\mathrm{supp}(f)|$, where $\mathrm{supp}(f)$ is the spectral support of f, that is, the set of coordinates S for which $\hat{f}(S) \neq 0$.

By $W(f)$, we denote the *total energy* of f, which is the square of its spectral 2-norm. For a collection $\mathcal{S} \subseteq 2^{[d]}$, the energy of f concentrated on $\mathcal{S}$ is given by $W_{\mathcal{S}}(f) = \sum_{S \in \mathcal{S}} \hat{f}(S)^2$. The *influence* of variable i on f, denoted $I_i[f]$, is the energy of f concentrated on the set $\{S : i \in S\}$, and the *restricted influence* up to degree k, denoted $R_i^k[f]$, is the energy of f concentrated on $\{S : i \in S, |S| \leq k\}$.

Finally, for a feature subset $K \subseteq [d]$, the *projection* of a Boolean function f onto K is the pseudo-Boolean function f_K defined by

$$f_K(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{z} \sim \mathcal{U}}[f(\boldsymbol{z}) \mid \boldsymbol{z}_K = \boldsymbol{x}_K] = \sum_{S \subseteq K} \hat{f}(S)\phi_S(\boldsymbol{x}).$$

Note that while f is Boolean, its projection f_K is generally pseudo-Boolean with values in $[-1, 1]$. Its total energy $W(f_K)$ corresponds to the energy of f concentrated on the lattice 2^K. The following identity establishes the fundamental link between the spectral energy of the projection and the global relevance error.

Lemma 1. For any Boolean function f and any subset $K \subseteq [d]$, the global relevance error under the uniform distribution $\mathcal{D} = \mathcal{U} \times \mathcal{U}$ is given by:

$$\mathsf{err}_{f,\mathcal{D}}(K) = \frac{1}{2}\left(1 - W(f_K)\right).$$

3.1 Spectral Characterization of Relevance

In the context where $\mathcal{P}$ is the uniform distribution over instance pairs, the following lemma states that minimizing the relevance error of a subset K with respect to a classifier f is equivalent to maximizing the energy of the projected function f_K.

Lemma 2. Let f be a d-dimensional Boolean function and let $\mathcal{P} = \mathcal{U} \times \mathcal{U}$ be the uniform distribution over pairs. For any $K \subseteq [d]$, the relevance error of K with respect to f and $\mathcal{P}$ satisfies:

$$\mathsf{err}_{f,\mathcal{P}}(K) = \frac{1}{2} - \frac{1}{2}W(f_K). \tag{3}$$

Proof. In what follows, let $\mathcal{U}_K$ denote the uniform distribution over $\{0,1\}^{|K|}$. Using the identity $\mathbb{P}[A \neq B] = \frac{1}{2} - \frac{1}{2}\mathbb{E}[AB]$ for random variables $A, B \in \{-1, +1\}$, we have:

$$\mathsf{err}_{f,\mathcal{P}}(K) = \frac{1}{2} - \frac{1}{2}\mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim\mathcal{P}}[f(\boldsymbol{x})f(\boldsymbol{z}) \mid \boldsymbol{x}_K = \boldsymbol{z}_K].$$

By the law of total expectation, we condition on the specific assignment $\boldsymbol{a} \in \{0,1\}^{|K|}$ shared by the features in K:

$$\begin{aligned}\mathbb{E}[f(\boldsymbol{x})f(\boldsymbol{z}) \mid \boldsymbol{x}_K = \boldsymbol{z}_K] &= \mathbb{E}_{\boldsymbol{a}\sim\mathcal{U}_K}[\mathbb{E}[f(\boldsymbol{x})f(\boldsymbol{z}) \mid \boldsymbol{x}_K = \boldsymbol{a}, \boldsymbol{z}_K = \boldsymbol{a}]] \\ &= \mathbb{E}_{\boldsymbol{a}\sim\mathcal{U}_K}[\mathbb{E}[f(\boldsymbol{x}) \mid \boldsymbol{x}_K = \boldsymbol{a}] \cdot \mathbb{E}[f(\boldsymbol{z}) \mid \boldsymbol{z}_K = \boldsymbol{a}]] \\ &= \mathbb{E}_{\boldsymbol{a}\sim\mathcal{U}_K}[(f_K(\boldsymbol{a}))^2] = \|f_K\|_2^2.\end{aligned}$$

The second equality relies on the independence of $\boldsymbol{x}$ and $\boldsymbol{z}$ under $\mathcal{P}$. By Parseval's identity, we know that $\|f_K\|_2^2 = \sum_{S\subseteq K} \hat{f}(S)^2 = W(f_K)$. Substituting this energy term into the error expression yields the desired identity (3). □

3.2 Spectral Recovery of Projections

When f is an arbitrary Boolean function only accessible through membership queries, an exact computation of the Fourier spectrum is generally infeasible. However, the Kushilevitz-Mansour (KM) algorithm [9] allows us to efficiently identify a pseudo-Boolean function g that approximates f in the L_∞ norm and possesses a sparse Fourier spectrum. Its properties are summarized in the following claim.

Claim 1. There exists an algorithm such that for any d-dimensional Boolean function f, given a threshold $\theta \in (0,1)$, a confidence parameter $\delta \in (0,1)$, and membership query access to f, the algorithm runs in time $\mathrm{poly}(d, 1/\theta, \ln(1/\delta))$ and outputs the Fourier expansion of a pseudo-Boolean function g such that, with probability at least $1-\delta$:

$$L_\infty(f-g) \leq \theta \quad \text{and} \quad L_0(g) \leq \frac{4}{\theta^2}.$$

The fact that we can efficiently approximate f with a function g that is close in terms of coefficient magnitude allows us to bound the difference between the energies of their projections onto any subset K of size at most k. This is formalized in the following lemma.

Lemma 3. Let f be a d-dimensional Boolean function and let g be a pseudo-Boolean function such that $L_\infty(f-g) \leq \epsilon/2^k$ for some $\epsilon \in (0,1)$. Then, for any subset $K \subseteq [d]$ of size at most k, the difference between their projected energies satisfies:

$$|W(f_K) - W(g_K)| \leq 3\epsilon.$$

Proof. By applying the difference of squares, for any $S \subseteq K$, we have:

$$|\hat{f}(S)^2 - \hat{g}(S)^2| = |(\hat{f}(S) - \hat{g}(S))(\hat{f}(S) + \hat{g}(S))| \leq |\hat{f}(S) - \hat{g}(S)| \cdot (|\hat{f}(S)| + |\hat{g}(S)|).$$

Let $\theta = \epsilon/2^k$. Since $L_\infty(f-g) \leq \theta$, it holds that $|\hat{f}(S) - \hat{g}(S)| \leq \theta$. Furthermore, as f is Boolean, $|\hat{f}(S)| \leq 1$, which implies $|\hat{g}(S)| \leq 1 + \theta$. Substituting these bounds yields:

$$|\hat{f}(S)^2 - \hat{g}(S)^2| \leq \theta \cdot (2 + \theta) = 2\theta + \theta^2.$$

Summing over all $2^{|K|}$ subsets $S \subseteq K$ and applying the triangle inequality:

$$|W(f_K) - W(g_K)| \leq \sum_{S \subseteq K} |\hat{f}(S)^2 - \hat{g}(S)^2| \leq 2^k(2\theta + \theta^2).$$

Substituting $\theta = \epsilon/2^k$, we obtain $2\epsilon + \epsilon^2/2^k$. Since $\epsilon < 1$ and $2^k \geq 1$, we have $2\epsilon + \epsilon^2/2^k \leq 3\epsilon$, completing the proof. □

Thus, if we choose a threshold θ that is sufficiently small relative to the size limit k and the desired accuracy ϵ, we can ensure that g serves as a reliable proxy for f. Specifically, maximizing $W(g_K)$ over subsets of size k provides a near-optimal solution for minimizing the global relevance error of f.

3.3 Feature Pruning of Projections

Although the function g returned by the KM algorithm is sparse, its spectral support $\mathrm{supp}(g)$ may still involve a large number of distinct variables, potentially leading to an exhaustive search over $O(d^k)$ subsets. To address this, we apply a pruning strategy inspired by Gopalan, Kalai, and Khanna [6] to identify a restricted set of influential features that captures the bulk of the energy of g_K.

Lemma 4. Let g be a d-dimensional pseudo-Boolean function, and let $I = \{i \in [d] : R_i^k[g] \geq \epsilon/k\}$ be the set of features with significant restricted influence for some $\epsilon \in (0, 1)$. Let h denote the projection of g onto I. Then, for any subset $K \subseteq [d]$ of size at most k, the energy of g on K is bounded by:

$$W(g_K) \leq W(h_{K \cap I}) + \epsilon.$$

Proof. Let $\mathcal{K}_I = \{S \subseteq K : S \subseteq I\}$ be the collection of subsets of K that are entirely contained within the influential set I. We decompose the energy of g on K as follows:

$$W(g_K) = \sum_{S \in \mathcal{K}_I} \hat{g}(S)^2 + \sum_{S \subseteq K, S \not\subseteq I} \hat{g}(S)^2.$$

The first term is precisely $W(h_{K \cap I})$ by the definition of h. For the second term, observe that any $S \subseteq K$ such that $S \not\subseteq I$ must contain at least one variable $i \in K \setminus I$. By the union bound on restricted influences:

$$\begin{aligned}\sum_{S \subseteq K, S \not\subseteq I} \hat{g}(S)^2 &\leq \sum_{i \in K \setminus I} \left(\sum_{S : i \in S, S \subseteq K} \hat{g}(S)^2 \right) \\ &\leq \sum_{i \in K \setminus I} R_i^k[g] < \sum_{i \in K \setminus I} \frac{\epsilon}{k} = |K \setminus I| \cdot \frac{\epsilon}{k} \leq \epsilon,\end{aligned}$$

where the last inequality follows from $|K| \leq k$. Combining these terms yields the desired bound. □

3.4 Main Result

Algorithm 1 implements the spectral approach to global explainability: it first leverages the KM algorithm to identify a sparse pseudo-Boolean function g that approximates f. It then prunes the feature space to a candidate set I of influential variables and performs an exhaustive search within this restricted space. As stated by the following theorem, this algorithm is fixed-parameter tractable in k and $1/\epsilon$, and achieves near-optimal relevance with high probability.

Theorem 3. For the uniform distribution $\mathcal{P} = \mathcal{U} \times \mathcal{U}$, there exists an algorithm such that for any d-dimensional Boolean function f, given a size constraint $k \leq d$, a confidence $\delta \in (0, 1)$, and an error $\epsilon \in (0, 1)$, along with membership query access to f, the algorithm returns a subset $K \subseteq [d]$ of size at most k such that with probability at least $1 - \delta$:

$$\mathsf{err}_{f,\mathcal{P}}(K) \leq \min_{K' \in \mathcal{K}_{d,k}} \mathsf{err}_{f,\mathcal{P}}(K') + \epsilon.$$

The algorithm runs in time $\mathrm{poly}(d, (k/\epsilon)^k, \ln(1/\delta))$.

Algorithm 1: Global Probabilistic Explainer (GPE)

Input: Membership query access to Boolean function f, size limit k, accuracy ϵ, confidence δ.

`// Spectral Recovery`
Run KM on f with threshold $\theta = \epsilon/(7 \cdot 2^k)$ and confidence δ to obtain pseudo-Boolean function g
`// Feature Pruning`
Identify the set of influential features $I = \{i \in [d] : R_i^k[g] \geq \epsilon/k\}$ and let h be the projection of g onto I
`// Subset Selection`
Find $K \subseteq I$ with $|K| \leq k$ that maximizes the projected energy $W(h_K)$
return K

Proof. Let K^* be a subset of size at most k that maximizes the projected energy $W(f_K)$. By Lemma 2, it suffices to show that the subset K returned by Algorithm 1 satisfies $W(f_K) \geq W(f_{K^*}) - 2\epsilon$. Assume the KM phase succeeds, which occurs with probability at least $1 - \delta$. Let $\theta = \epsilon/(7 \cdot 2^k)$. By Lemma 3, we have $|W(f_{K^*}) - W(g_{K^*})| \leq 3(\epsilon/7)$. Similarly, by Lemma 4, we have $W(g_{K^*}) \leq W(h_{K^* \cap I}) + \epsilon/7$. Since K is the maximizer of $W(h_K)$ over subsets of I, and $K^* \cap I$ is a candidate, we have $W(h_K) \geq W(h_{K^* \cap I})$.

Note that for any $K \subseteq I$, $W(h_K) = W(g_K)$. Combining the inequalities:

$$\begin{aligned} W(f_K) &\geq W(g_K) - \frac{3\epsilon}{7} = W(h_K) - \frac{3\epsilon}{7} \geq W(h_{K^* \cap I}) - \frac{3\epsilon}{7} \\ &\geq \left(W(g_{K^*}) - \frac{\epsilon}{7}\right) - \frac{3\epsilon}{7} \geq \left(W(f_{K^*}) - \frac{3\epsilon}{7}\right) - \frac{4\epsilon}{7} = W(f_{K^*}) - \epsilon. \end{aligned}$$

This confirms that $W(f_K)$ is within ϵ of the optimum energy, which corresponds to the ϵ-error bound in relevance.

For the complexity, the KM phase is $\mathrm{poly}(d, 2^k/\epsilon)$. The pruning phase identifies I, where $|I| \leq \sum R_i^k[g]/(\epsilon/k) \leq k^2/\epsilon$. Searching $\binom{|I|}{k}$ subsets takes $O((ek/\epsilon)^k)$ time. The total runtime is thus $\mathrm{poly}(d, (k/\epsilon)^k, \ln(1/\delta))$. □

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